7. In a polar coordinate system, O is the pole. The polar coordinates of the points P, Q and R are $(11, 59°)$, $(60, 149°)$ and $(144, 239°)$ respectively.

(a) Find $\angle POQ$.

(b) Are P, O and R collinear? Explain your answer.

(c) Find the perimeter of ΔPQR.

(4 marks)

(a) $\angle POQ = 149° - 59°$

$= 90°$

(b) $\angle POR = 239° - 59°$

$= 180°$

$\therefore$ P, O and R are collinear

(c) $PQ = \sqrt{11^2 + 60^2} = 61$

$QR = \sqrt{60^2 + 144^2} = 156$

$\therefore$ Perimeter $\Delta PQR = 61 + 156 + (11 + 144)$

$= 372$

Revision Exercise for 2nd Term Test

Selected Past Paper Questions

Amendments are made on p.6, 7, 9

Question

Q.1: 1617 1st Exam

2. The following table lists the atomic numbers of some elements in ✻ Group I of the periodic table.

Element	Atomic number
Li	3
Na	11
K	19
Rb	37

(a) (i) What is the special name of this group? (1 mark)

Group I = Alkali Metal

(ii) State ONE physical property of this group. (1 mark)

Shiny / silvery

(b) In what way are the electronic arrangements of Na and Rb

(i) similar to each other ? (1 mark)

outermost shell electron = 1

(ii) different from each other ? (1 mark)

Different periods = diffrent electron shells.

(c) When Na and Rb are dropped to water respectively, an alkali solution is formed.

Metal (Alkali) + Water → solution (Alkali)

(i) How would you show that an alkali is formed? (2 marks)

pH paper

(ii) State the observations between Na and water (3 marks)

✻ sodium is lighter than water

(iv) State which element, Na or Rb, will react more vigorously with water?

Explain your answer. (2 marks)

Rb ✓

More electrons.

= More unstable.